

			generator and battery.	
8	(a)	(i)	P = 236, Q = 92 R = 143	B1 B1
		(ii)	E = (Δm) c ² = [(235.1 + 1.009) – (148.0 + 84.9 + 3 x 1.009)] u c ² = (0.182) (1.66 x 10 ⁻²⁷) (3.00 x 10 ⁸) ² = 2.719 x 10 ⁻¹¹ J for only one uranium nucleus	M1 A1
	(b)	(i)	It is <u>not</u> triggered or affected by external conditions (temperature, pressure, et cetera)	B1

			It is impossible to predict which nucleus will decay at a particular instant or when a particular nucleus will decay	B1
		(ii)	${}_{38}^{90}\text{Sr} \rightarrow {}_{39}^{90}\text{Y} + {}_{-1}^0\beta$ Both correct yttrium and beta numbers	B1
		(iii)	$N = N_0 e^{-\lambda t} \text{ where } \lambda = \frac{\ln 2}{t_{1/2}} \text{ (equations provided in data sheet)}$ $= (2.36 \times 10^{13}) e^{-\left(\frac{\ln 2}{28}\right)(112)}$ $= (2.36 \times 10^{13})(0.0625)$ $= 1.48 \times 10^{12}$ <u>Alternative</u> Half life is 28 years so 112 years is 4 half lives Number present after this time is 1/16 of original Number present = $2.36 \times 10^{13} / 16 = 1.47(5) \times 10^{12}$	M0 M1 A1

Questions			Answers	Marks
2	(a)		Using $E = mc^2$, we get $1.70 \times 10^4 = m \times 5100 \times (50.0 - 10.0)$	